AWS IAM — Important Interview Questions & Answers

1.What is IAM?
IAM stands for Identity and Access Management. It is an AWS service used to securely control who can access AWS resources and what actions they are allowed to perform.
Simple ga gurthupettuko:
IAM ante AWS resources ki evariki access undali and em actions cheyyalo control chese service.
Eng: IAM is an AWS service used to control access to AWS resources.

2.What is an IAM User?
An IAM User is an identity created in AWS for a person or application that needs to access AWS resources. An IAM User can have permissions through policies.
Simple ga gurthupettuko:
IAM User ante AWS resources access cheyyadaniki create chese individual identity.
Eng: An IAM User is an identity used to access AWS resources.

3.What is an IAM Group?
An IAM Group is a collection of IAM Users. We can attach policies to a group so that all users in the group receive the required permissions.
Simple ga gurthupettuko:
IAM Group ante multiple IAM Users ni group ga manage chese place.
Eng: An IAM Group is a collection of IAM Users.

4.What is an IAM Role?
An IAM Role is an identity that provides temporary permissions to AWS resources or trusted entities. It is commonly used when an EC2 instance, Lambda function, or another AWS service needs access to resources.
Simple ga gurthupettuko:
IAM Role ante temporary permissions provide chesi AWS resources ki access ivvadaniki use chestham.
Eng: An IAM Role provides temporary permissions to trusted entities and AWS resources.

5.What is an IAM Policy?
An IAM Policy is a document that defines what actions are allowed or denied on AWS resources. Policies contain permissions that can be attached to users, groups, or roles.
Simple ga gurthupettuko:
IAM Policy ante evaru em action cheyyalo allow or deny chese permissions document.
Eng: An IAM Policy defines which actions are allowed or denied on AWS resources.

6.What is MFA in IAM?
MFA stands for Multi-Factor Authentication. It adds an extra layer of security by requiring an additional authentication factor along with the password.
Simple ga gurthupettuko:
MFA ante password tho patu additional verification use chesi account security improve cheyyadam.
Eng: MFA adds an extra layer of security to IAM authentication.

7.What are Access Keys?
Access Keys are credentials used to make programmatic requests to AWS services. They normally consist of an Access Key ID and a Secret Access Key.
Simple ga gurthupettuko:
Access Keys ante CLI, SDK, or applications dwara AWS services ni access cheyyadaniki use chese credentials.
Eng: Access Keys are credentials used for programmatic access to AWS services.

8.What are Permission Boundaries?
Permission Boundaries define the maximum permissions that an IAM User or IAM Role can have. They do not directly grant permissions but limit the maximum permissions that can be used.
Simple ga gurthupettuko:
Permission Boundary ante User or Role ki maximum ga entha permission undalo limit chese setting.
Eng: Permission Boundaries define the maximum permissions an IAM identity can have.

9.What is the difference between IAM User and IAM Role?
An IAM User is a long-term identity usually associated with a person or application, while an IAM Role provides temporary

credentials to trusted entities.
Simple ga gurthupettuko:
IAM User long-term identity, IAM Role temporary access kosam use chestham.
Eng: IAM Users are identities, while IAM Roles provide temporary access.

10.What is the difference between IAM User and IAM Group?
An IAM User represents an individual identity, while an IAM Group is a collection of IAM Users. Policies can be attached to groups to manage permissions for multiple users.
Simple ga gurthupettuko:
User ante individual identity, Group ante multiple users collection.
Eng: An IAM User is an individual identity, while an IAM Group is a collection of users.

11.What is the difference between IAM Policy and Permission Boundary?
An IAM Policy grants or denies permissions to an identity, while a Permission Boundary defines the maximum permissions that the identity can use.
Simple ga gurthupettuko:
Policy permissions grant or deny chestundi, Permission Boundary maximum permission ni limit chestundi.
Eng: A Policy grants permissions, while a Permission Boundary limits maximum permissions.

12.What is Authentication in IAM?
Authentication is the process of verifying who a user or identity is. In AWS, authentication can be done using passwords, MFA, access keys, or other supported authentication methods.
Simple ga gurthupettuko:
Authentication ante user evaru ani verify cheyyadam.
Eng: Authentication is the process of verifying an identity.

13.What is Authorization in IAM?
Authorization is the process of determining what actions an authenticated identity is allowed or denied to perform on AWS resources.
Simple ga gurthupettuko:
Authorization ante authenticated user em actions cheyyagalado decide cheyyadam.
Eng: Authorization determines what actions an identity is allowed to perform.

14.What is the Principle of Least Privilege?
The Principle of Least Privilege means giving users, roles, and applications only the permissions they actually need to perform their tasks.
Simple ga gurthupettuko:
Least Privilege ante required permissions matrame ivvali, extra permissions ivvakudadhu.
Eng: Least Privilege means giving only the permissions that are required.

15.Can an IAM User belong to multiple groups?
Yes. An IAM User can belong to multiple IAM Groups. The permissions from the groups can be combined with the user's other applicable permissions.
Simple ga gurthupettuko:
Oka IAM User multiple groups lo member ga undachu.
Eng: An IAM User can belong to multiple IAM Groups.

16.Can an IAM Role be assigned to an EC2 instance?
Yes. An IAM Role can be attached to an EC2 instance through an instance profile. This allows the EC2 instance to obtain temporary credentials and access permitted AWS resources.
Simple ga gurthupettuko:
EC2 ki IAM Role attach chesi AWS resources ki secure temporary access ivvachu.
Eng: An IAM Role can be attached to an EC2 instance through an instance profile.

17.Why do we use IAM Roles instead of Access Keys for EC2?
IAM Roles provide temporary credentials to EC2 instances and avoid storing long-term access keys on the server. This is generally a more secure approach for AWS resource access.
Simple ga gurthupettuko:
EC2 lo Access Keys store cheyyakunda IAM Role use chesthe secure temporary access vastundi.
Eng: IAM Roles provide temporary credentials and avoid storing long-term access keys on EC2.

18.What happens if an IAM User has multiple policies?
The permissions from applicable policies are evaluated together. An explicit Deny overrides an Allow when determining whether an action is permitted.
Simple ga gurthupettuko:
Multiple policies unte permissions combine avuthayi, kani explicit Deny unte Allow ni override chestundi.
Eng: Multiple policies are evaluated together, and an explicit Deny overrides an Allow.

19.What is an Explicit Deny?
An Explicit Deny is a policy statement that specifically denies an action. If an applicable policy contains an explicit Deny, it overrides any Allow for that action.
Simple ga gurthupettuko:
Explicit Deny ante specific action ni direct ga deny cheyyadam, idi Allow kanna powerful.
Eng: An Explicit Deny overrides an Allow for the same action.

20.How does IAM control access to AWS resources?
IAM uses identities, policies, roles, and permissions to determine whether an action is allowed or denied. AWS evaluates the applicable policies before allowing access to a resource.
Simple ga gurthupettuko:
IAM policies and permissions ni evaluate chesi AWS resource access allow or deny chestundi.
Eng: IAM controls access by evaluating applicable policies and permissions.

21.What is the difference between Authentication and Authorization?
Authentication verifies who the user is, while Authorization determines what the authenticated user is allowed to do.
Simple ga gurthupettuko:
Authentication ante evaru ani verify cheyyadam, Authorization ante em cheyyalo decide cheyyadam.
Eng: Authentication verifies identity, while Authorization determines permissions.

22.How would you secure an IAM User?
I would enable MFA, follow the Principle of Least Privilege, avoid unnecessary access keys, use strong authentication, and regularly review permissions and credentials.
Simple ga gurthupettuko:
IAM User security kosam MFA enable chesi required permissions matrame ivvali.
Eng: I would use MFA, least privilege, and regular permission reviews to secure an IAM User.

23.How would you troubleshoot an IAM Access Denied error?
I would check the IAM policy, resource policy if applicable, explicit denies, permission boundaries, identity permissions, and the requested action and resource. I would also verify whether the correct role or user is being used.
Simple ga gurthupettuko:
Access Denied vaste Policy → Explicit Deny → Permission Boundary → Resource Policy → User/Role check chestham.
Eng: For an IAM Access Denied error, I would check policies, explicit denies, boundaries, and the identity being used.

24.What is the most important thing to remember about IAM?
IAM is used to securely manage identities and permissions in AWS. The important concepts are Users, Groups, Roles, Policies, MFA, Access Keys, and Permission Boundaries.
Simple ga gurthupettuko:
IAM ante AWS resources ki secure access and permissions manage cheyyadaniki use chese service.
Eng: IAM is used to securely manage identities and permissions in AWS.